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CHE 313 Transport Phenomena III

Quiz #1 (1/30/20)

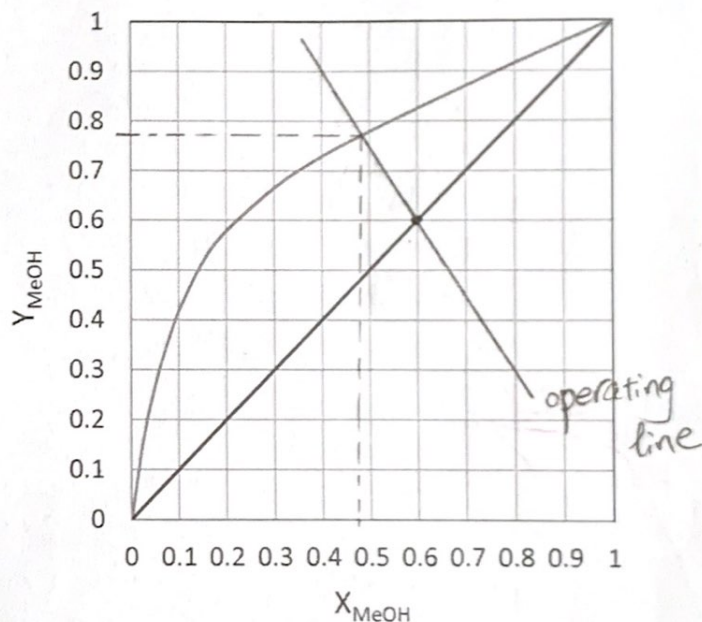
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1. We are separating a mixture of methanol and water in a flash drum at 1.0 atm pressure. Feed is 60.0 mol% methanol, and 40.0 % of the feed is vaporized. Feed flow rate is 100 kmol/hr. $z = 0.6$

$f = \frac{V}{F} = 0.4$
 $V = 0.4 F = 40$ $L = 60$

Equilibrium data and xy graph are:

X_methanol	Y_methanol	Temp, C
0	0	100
0.02	0.134	96.4
0.04	0.23	93.5
0.06	0.304	91.2
0.08	0.365	89.3
0.1	0.418	87.7
0.15	0.517	84.4
0.2	0.579	81.7
0.3	0.665	78
0.4	0.729	75.3
0.5	0.779	73.1
0.6	0.825	71.2
0.7	0.87	69.3
0.8	0.915	67.6
0.9	0.958	66
0.95	0.979	65
1	1	64.5



$\frac{L}{V}x + \frac{F}{V}z$

- a) Find a slope of the operating line and a point on $y=x$ curve using overall mass balance and component mass balance equations. (hint: $y=x=z$) $z=0.6$

slope = $-\frac{L}{V} = -\frac{60}{40} = -\frac{3}{2}$ ✓ point = $(0.6, 0.6)$ ✓

- b) Draw the operating line on the graph. $y = -\frac{3}{2}x + \frac{100}{40} \cdot (0.6) \Rightarrow y = -\frac{3}{2}x + 1.5$ ✓

- c) Find the vapor (y) and liquid (x) mole fractions, liquid flow rate (L), and vapor flow rate (V).

$y_{\text{meth}} \approx 0.78$ $y_{\text{water}} = 1 - y_{\text{meth}} = 0.22$
 $x_{\text{meth}} \approx 0.48$ $x_{\text{water}} = 0.52$ ✓

$f = \frac{V}{F} = 0.4$

$V = 0.4 F = 40 \frac{\text{kmol}}{\text{hr}}$

$F = V + L \Rightarrow L = F - V = 60 \frac{\text{kmol}}{\text{hr}}$

$$RR = 2 \frac{(k_i - 1) z_i}{1 + (k_i - 1) \frac{V}{F}}$$

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2. We have a liquid feed that is 20 mol% methane, 45 mol% n-pentane, and 35 mol% n-hexane. Feed rate is 1500 kmol/h, feed temperature is 45°C, and pressure is 100.0 psia. The flash drum operates at 30 psia and is adiabatic.

In the class, we practiced the double trial and error (simultaneous multicomponent convergence) procedures when T_{drum} is unknown. Make your own flowsheet to find T_{drum} , V/F , x_i , y_i , L , and V .

- 1) start at $T = 15^\circ\text{C}$ (drum)
- 2) Based on T_{drum} & P_{drum} we have, find k_i from D-P chart
- 3) using R-R eq and the k_i , solve for $\frac{V}{F}$ by setting $RR = 0$
- 4) once we have $\frac{V}{F}$, check if that satisfies

$$F = V + L$$

$$F \cdot z_i = V \cdot y_i + L \cdot x_i$$

should move to here!

if yes

5.a) check if satisfies

$$F \cdot h_f + Q_f = V \cdot h_v + L \cdot h_L$$

6.a) if yes

check if $RR < 0.005$

answer

calculate:

$$x_i = \frac{z_i}{1 + (k_i - 1) \frac{V}{F}}$$

$$y_i = k_i \cdot x_i$$

$$V = \frac{V}{F} \cdot F$$

$$L = F - V$$

if no

$$5.b) \frac{V}{F}_{\text{new}} = \frac{V}{F}_{\text{old}} + \frac{f(\frac{V}{F})}{\frac{df(\frac{V}{F})}{d(\frac{V}{F})}}$$

then repeat

step 4) until $RR < 0.005$

$$T_{\text{new}} = T_{\text{old}} + \frac{T_{\text{old}}}{\frac{dE}{d(T)}}$$

repeat step 1